Classifying Autism from Resting-State fMRI: Max-Min vs Correlation Features with Logistic Regression and SVM

UE24CS352A Machine Learning - Mini-Project Write-up
Team: [Jayanth K,PES1UG24AM123| [Pradeep R,PES1UG24AM809] | Problem no.: [30] | Repo: [https://github.com/jayanthh02/MachineLearning-MiniProject]

1. Problem Statement

We follow the pipeline of *Classifying Adolescent Excessive Alcohol Drinkers from fMRI Data* (Kim, Liu and Noh, Stanford CS229), which predicts a binary label from resting-state fMRI signals using Logistic Regression (LR), SVMs and deep models. Its NCANDA data is access-restricted, so, with our instructor's approval, we applied the same pipeline to the public ABIDE dataset: **classify autism vs typical control** from resting-state fMRI. The paper summarises each brain region by its signal range (max-min). We ask whether a richer feature, the **correlation between pairs of regions**, does better, and whether demographics (age, sex, scanner site) explain the results.

2. Dataset

ABIDE I (preprocessed, CPAC pipeline, no filtering, no global signal regression), loaded with `nilearn.datasets.fetch_abide_pcp`. It has **871 subjects: 468 typical controls and 403 autism**, so always predicting 'control' gives a baseline accuracy of **0.537**. The data comes from 20 scanning sites; the number of time points T per subject varies from 78 to 316. Each subject is a T x N table: rows are fMRI snapshots, columns are brain regions (BOLD signal averaged over each region). We use two ready-made parcellations: **Craddock-200** (N = 200) and **Harvard-Oxford** (N = 111). The paper's ICA parcellation was not available as a ready-made file, so Harvard-Oxford is our second atlas. Demographics used: age (no missing values), sex, and site (one-hot, 20 columns).

3. Approach

- **Feature set A (from the paper): max-min.** Z-score each region's time series, then keep max - min per region. This collapses T and gives N features per subject (200 or 111).
- **Feature set B (our modification): correlation.** Compute the Pearson correlation between every pair of regions, keep the upper triangle, and apply the Fisher z-transform. This gives N(N-1)/2 features: 19,900 (Craddock) or 6,105 (Harvard-Oxford). Max-min measures how much each region swings on its own; correlation measures which regions rise and fall *together*, which max-min cannot see.
- **Models (syllabus):** Logistic Regression and SVM with linear, polynomial (degree 2), RBF and sigmoid kernels.
- **Demographic settings:** brain only, brain + demographics, brain + demographics without age, and demographics only (control).
- **Evaluation (as in the paper):** a stratified 15% dev set (131 subjects) to choose the regularisation parameter C; 10-fold stratified cross-validation on the other 740 subjects for reported accuracy. Fixed random seed.

4. Implementation Overview

Python with nilearn, NumPy, pandas and scikit-learn, organised as four stages in one Jupyter notebook: (1) download ABIDE (cached locally), (2) build the four feature matrices and the demographics table, (3) LR across all feature sets and settings, (4) SVMs. For LR, a `StandardScaler` sits inside a scikit-learn pipeline, so it is fitted on training folds only and nothing leaks from the test fold. For SVMs on correlation features, fitting on ~20,000 raw features took about 12 minutes per model, so we compute each 871 x 871 kernel matrix once and train with `kernel='precomputed'`, which is far faster. These SVM runs use no fitted scaler, so they are not exactly comparable to the scaled LR runs.

5. Results

10-fold CV accuracy, brain features only. Baseline (majority class) = 0.537. The standard deviation across folds is about 0.06, so the standard error of each mean is about 0.02.

Features	LR	SVM linear	SVM poly2	SVM RBF	SVM sigmoid
Craddock-200, max-min	0.541	0.515	0.541	0.516	0.538
Craddock-200, correlation	0.666	0.659	0.703	0.677	0.538
Harvard-Oxford, max-min	0.485	0.473	0.469	0.535	n/a
Harvard-Oxford, correlation	0.650	0.680	0.658	0.642	0.538

Demographics only (LR): 0.530. Adding demographics to the brain features changed LR accuracy by under 0.01 on the correlation features (e.g. Craddock 0.666 to 0.665; Harvard-Oxford unchanged at 0.650). Harvard-Oxford max-min with the sigmoid kernel was not recorded (n/a).

6. Conclusions

- **Correlation features clearly beat the paper's max-min features:** about 0.64 to 0.70 vs 0.47 to 0.54, about 10 to 17 points above the 0.537 baseline, and LR and SVM agree. Keeping how regions move together is what carries the signal; the size of each region's swing carries almost none.
- **Demographics predict almost nothing here** (0.530 alone), unlike the paper, where age drove the result. Because only 22 demographic columns sit among thousands of brain features and are heavily regularised, the demographics-only result is the more reliable evidence than the with/without-age rows.
- **No kernel wins consistently:** poly2 is best on Craddock (0.703) but linear is best on Harvard-Oxford (0.680); gaps of 2 to 4 points are within noise. The **sigmoid kernel failed to learn** (it predicted the majority class), as the paper also saw.
- **Limitations:** the dataset differs from the paper's, so numbers are not directly comparable; C was chosen on a small dev set; Harvard-Oxford replaced ICA; and scanner (site) effects were not tested with leave-one-site-out validation. Future work: leave-site-out testing and additional atlases.

Reference: Y. Kim, C. Liu, J. Noh, *Classifying Adolescent Excessive Alcohol Drinkers from fMRI Data*, Stanford CS229 project report.